Reference Event 1: Coordinated Cyber Attack on Minnesota Community Water Systems

Characterization Against the Water Sector Design Basis

DBA-W-RE-1 — Reference Event Instrument, DBA-W Series

Lineage: Issued under the DBA-W Sector Framework v1.5; DBA-MA Parent Framework

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This version is a member of CIKR Design Basis Examples, Edition 1 (November 2026) (https://doi.org/10.5281/zenodo.23863986), a dated edition that freezes, for each of fifteen critical infrastructure sectors, a sector framework, a facility class and a reference event as they stood on 1 November 2026. The edition is not falsified by a later revision of this document, whose working master continues to move. It is a working draft, marked DRAFT on its cover.

1. Purpose, Scope and Position

1.1 Purpose

This instrument reads the coordinated cyber attack on Minnesota community water systems of 26–27 July 2026, and the multi-state campaign of which it was the first-reported part, back through the Water Sector Design Basis. It returns a tier determination (Section 5), a reading of each provision the event exercised (Section 6), eight findings and ten recommendations. It was the first reference event instrument in the DBA series, and the model to which the later instruments were written.

1.2 Position and Scope

A reference event reasons backward from an event that occurred and asks whether the methodology, as written, was sufficient; a use case reasons forward. This instrument recommends and does not amend; leaves the applicability problem of F-5 to OI-2; and treats consequence within the water system boundary only. The Bright Line resolves to FD-BL (*The Bright Line*) and its determinations FD-BL-D1 (*Mode of Discharge*) and FD-BL-D2 (*Indication Integrity*), envelope terminology to FD-EV (*The Envelope*); the Foundational Definitions are cited by designation, FD-REG holding their versions. Section references of the bare form §N are to the DBA-W Sector Framework v1.5, the version this instrument read. The §9.3 thresholds that v1.5 calls Bright Line criteria are called coordination thresholds here, as the framework now calls them, the term being reserved to FD-BL.

2. Sources and Attribution

2.1 Sources and Their Limit

Public sources only: advisory AA26-097A of the Cybersecurity and Infrastructure Security Agency, published 7 April 2026 and updated 22 July 2026; the Public Service Announcement of the Federal Bureau of Investigation and the Environmental Protection Agency of 30 July 2026, modified 6 August 2026, primary for the access condition, effects, cross-site detection and third-party configuration; Minnesota IT Services' statement of 28 July 2026; and vendor, research and press sources listed in the predecessor.

2.2 Evidentiary Classes

Confirmed: stated by an official body with authority over the matter, including an affected municipality speaking of its own systems. Reported: stated by a named non-official source with standing. Assessed: inference drawn here, marked as such.

2.3 Attribution Posture

No official attribution of the Minnesota events to any specific actor had been made at the time of writing; federal and state officials had not named a responsible party, and the investigation remained active. An Iran-aligned group known as CyberAv3ngers, the Islamic Revolutionary Guard Corps–linked actor associated with the 2023 Unitronics programmable-logic-controller campaign, has publicly claimed responsibility for the Minnesota events and characterized them as intended "only to warn." The claim is recorded as reported, not confirmed: no official body has adopted it, and a claim of responsibility is not an official attribution. Reported assessments hold that the operational pattern is consistent with the threat ecosystem documented in AA26-097A, which the United States Government has formally attributed to Iran's Islamic Revolutionary Guard Corps Cyber-Electronic Command.

No finding depends upon attribution: every finding follows from the decomposition in Section 4 and would stand if attribution were never established, or established differently. The tier determination of Section 5.5 is contingent on the observed characteristics, not on attribution (OI-1).

2.4 Currency of the Record

The record was developing at the time of writing. Section 3 is the state of public knowledge at issue; Sections 4 to 8 are the durable portion. The correction recorded at Section 9.8 (OI-8), that the wave beginning 27 July did not run through CVE-2021-22681, was reached independently about a month after this instrument made it: infracritical's *Water Sector Cyber Incident Response Playbook*, version 01A of 3 September 2026 (report IC-SCID-00150), retracts the same attribution on the same ground, that the vulnerability belongs to a different product family from the controllers involved. It is recorded as external corroboration of a correction already published; nothing else from that report enters this instrument.

3. The Event

3.1 Minnesota Community Water Systems, 26–27 July 2026

Between 26 and 27 July 2026, a coordinated cyber attack disrupted operational technology at more than thirty community water and wastewater systems across Minnesota, and Minnesota IT Services activated a statewide incident response. Four municipalities disclosed. In Braham, approximately 1,700 served, the attack disabled computerized controls and shut down the well and treatment plant; crews restored operation in approximately two hours. In Plymouth, approximately 80,000, the city disconnected cellular-connected equipment at two water towers and multiple wastewater lift stations and continued on manual procedures. South St. Paul and Maple Plain maintained service, Maple Plain declaring a local emergency. In every confirmed Minnesota case officials stated that drinking water quality was not affected.

3.1a Expansion Beyond Minnesota

By 4 August 2026, utilities in at least twelve states were reported to be remediating incidents attributed to the same campaign; Minnesota, Michigan, Georgia, New Jersey and South Dakota have been identified publicly. The Federal Bureau of Investigation stated that utilities in at least seven states had reported incidents since 27 July, some degrading water operations. Open-source analysis by the Center for Strategic and International Studies of 18 August 2026 placed the campaign at approximately one hundred affected facilities across at least twelve states, fifty-five geolocated and nine of the twelve having confirmed incidents publicly. The most consequential single incident is the Clayton County Water Authority's, near Atlanta, serving approximately 300,000: cyber activity caused a drop in water pressure and a precautionary boil water advisory, lifted after testing, with service restored within hours. The affected systems span the full range of size, drinking water and wastewater alike; F-6 is qualified accordingly.

3.2 The Advisory Campaign Context

AA26-097A documented actors exploiting internet-exposed programmable logic controllers in the Government Services, Water and Wastewater, and Energy sectors since at least March 2026. Its update of 22 July 2026, four days before the Minnesota events, widened targeting to Schneider Electric and Siemens controllers, recorded exfiltration of controller project files, and recorded the observation central here: at one victim a malicious project file kept normal downstream ladder logic while modified Add-On Instructions disabled safety shutdown and alarm systems, and data on supervisory displays was manipulated, so that equipment ran in unsafe conditions without alerting operators.

3.3 The Access Condition and Its Remediation Status

Two access conditions are documented. CVE-2021-22681, an authentication bypass in Rockwell Automation Logix controllers, disclosed in February 2021 and stated by the vendor to be not fully addressable by patch, is in the broader advisory record and is not, on the public evidence, the condition exploited here. The condition named in the announcement of 30 July 2026 governs the wave beginning 27 July: Rockwell Automation / Allen-Bradley MicroLogix 1100 and 1400 series controllers reachable from the internet, reconfigured remotely by changing IP addresses and enabling and setting passwords, producing loss of view and in some cases loss of function. The announcement names no vulnerability identifier and gives explicit attention to end-of-life hardware. Against the first condition there is no patch because the flaw cannot be patched; against the second, because the product line has ended; neither is resolvable within a program organized around patching (Section 6.4).

3.4 Exposure Surface

A scan of April 2026 found 5,219 internet-exposed hosts self-identifying as the primary manufacturer's devices, 3,891 of them in the United States, a disproportionate share on cellular

networks. Of 150,000 to 170,000 United States water systems, most are small, rural and without dedicated operational technology security staff, and the Environmental Protection Agency reported in 2024 that more than seventy percent were failing to comply with the risk assessment and emergency response plan provisions of America's Water Infrastructure Act of 2018.

3.5 Federal Response

On 30 July 2026 the Cybersecurity and Infrastructure Security Agency advised operators to remove water control technology from the public internet. The announcement of the same day records what the findings rest on: loss of pressure and flooding, pressure loss potentially allowing untreated ground water to seep into pipes (Section 6.9); modified project files found by noticing ladder logic discrepancies across several sites (Section 6.8); third-party configuration similarities that may let the actors multiply successes (Section 7.8); and impact dependent on the utility's capability to switch to manual operations. Neither the advisory nor the announcement distinguished by system size.

3.6 Geopolitical Frame

Recorded as reported context, on which no finding depends. The United States and Iran have been in active armed conflict since 28 February 2026, and Iranian cyber operations against United States infrastructure have been reported to escalate with it. It is further reported that the techniques of AA26-097A have proliferated to more than sixty affiliated groups; F-4 treats the consequence, which does not depend on the framing being accurate in its particulars.

4. Technical Decomposition

Four access, two effect and one consequence element, organized by the provision each engages rather than by adversary tactic, since the assessment needs what the facility must withstand. The elements are confirmed federal content; their combined effect (Section 4.3) and the fleet-scale reading of A-4 (Section 7.8) are assessed.

Table 4-1. Event elements and their mapping, to the DBA-W Sector Framework v1.5.

ID	Element	Provisions	Findings
A-1	Controllers reachable from the public internet, including field assets on cellular networks	§5.1; §4.2.2	None; Section 6.2
A-2	Software-replicable credential, no hardware binding, not resolvable by patch	§5.2; §4.2.2	None; Sections 6.1, 6.4
A-3	Legitimate engineering software run from leased adversary infrastructure	§9.1; §5.2	F-3; Section 6.8
E-1	Modified Add-On Instructions disable safety shutdown and alarms; ladder logic left normal	§4.1; §4.2.2; §5.2; §9.4	F-2, F-3; Sections 6.3, 6.7
E-2	Falsified indication on operator displays; unsafe operation without alert	§4.2.2; §4.3; §5.2	F-1; Sections 6.5, 6.11

A-4	Common configuration supplied by third parties across unrelated utilities	§4.1; §5.1; §9.3; §1	F-8; Section 7.8
C-1	Pressure loss and flooding; pressure loss a federally stated ground water intrusion pathway	§4.2; §4.3	R-8; Section 6.9

4.1 Access Elements

In A-2 the controller has no basis to tell an authorized change from an unauthorized one, and the adversary's configuration change is itself the lockout, converting access into denial of control in one step. A-3 is protocol-legitimate and tool-legitimate, distinguishable only organizationally: whether the workstation is authorized, the session in a maintenance window, and anyone at the utility initiated it.

4.2 Effect Elements

E-1 lies beneath the level at which programs are reviewed: the controller executes the process and has ceased to protect it. E-2 is not loss of visibility. An operator facing a dead display knows he is blind, and the manual backup of §5.2 is written for that; an operator facing falsified indication is confidently wrong. Loss of view occurred at most victims and falsified indication at one; the second is the limiting case and governs.

4.3 The Combination

E-1 and E-2 together remove the protective function and falsify the indication through one artifact, the controller project file: a single-substrate defeat of two layers the Design Basis treats as independent. The combination was not, on the public evidence, executed in Minnesota; it is federal observation at another victim of the same campaign, and because the inclusion criterion is consequence potential, the demonstrated technique is the reference condition.

5. Characterization Against the Four Military Action Characteristics

The four characteristics of §7.3 are read against the record.

5.1 Intentionality

Present. A device population and a sector were chosen and pursued over months. Assessed: the concentration of effect on small systems is consistent with selection for defensive weakness rather than consequence magnitude, a variant §7.3.1, which assumes selection for maximum consequence, does not address. The reported "only to warn" claim is consistent with this and does not alter Section 5.5.

5.2 Persistence

Present. Disruptions since at least March 2026 and actor infrastructure from September 2025 to July 2026; the unpatchable access condition makes persistence a property of the condition as well as the adversary.

5.3 Precision

Present, in the form §7.3.3 anticipates and more refined: specific modules within a controller program, chosen because they carry the protective function and are not visible where programs are reviewed.

5.4 Denial

Absent, structurally. Nothing prevented anyone reaching a facility: Braham's crews restored the plant in about two hours, Plymouth's reconfigured field equipment, and mutual aid and repair materials were available. Cyber effect at range cannot deny physical access in a permissive environment. E-2 denies the operator accurate knowledge, an input to correct action rather than access; F-1 treats that gap, and widening the characteristic to cover it would dissolve its distinction.

5.5 Determination

Three of the four characteristics are present and the fourth is structurally absent. The event does not constitute a military action tier event within the meaning of the DBA-MA extension. Under the partial implementation principle the four are addressed as a system: a three-of-four match is a different kind of event, not a weak military action event. The event occupies the upper bound of the adversarial tier: state-directed effect delivered without the military action envelope, where the cyber requirements of §5 are load-bearing. That is the tier determination returned for the DBA Reference Event Register; its re-examination conditions are OI-1.

6. Characterization Against the Design Basis

Section references are to the DBA-W Sector Framework v1.5. Each reading states its verdict first.

6.1 §5.2 Credential Architecture

Held. The requirement of a hardware authentication element not replicable through software, drafted before the campaign, prohibits by principle the condition A-2 exploits: the strongest validation in the assessment, and the advantage of requirements stated as architectural properties.

6.2 §5.1 The OT/IT Boundary

Held. A boundary that prevents unauthorized process modification by any network path reaches the cellular field asset that never crosses the enterprise network, and is better than the coarser federal advice to remove such systems from the internet.

6.3 §5.2 Physical Process Interlock Independence

Held, with a gap. The physical mode switch, which governs whether a controller accepts program modification, is an instance of the requirement, but the requirement does not name program modification, the technique of record. R-3.

6.4 §4.2.2 Architectural Rather Than Incident-Specific Protection

Held. The indefinite design basis duration, resting on architectural protection, survives an unpatchable condition against which a patch-based program had no path. Assessed: the most transferable argument for outreach.

6.5 §5.2 Incident Response and Manual Backup

Held, with a gap. Exercised in the field: Braham and Plymouth did what §5.2 requires. It holds against the Minnesota conditions and is incomplete against the demonstrated capability, because manual operation protects only while the operator's knowledge of process state is sound, and against E-2 it is not. The central negative result; F-1.

6.6 §9.3 Coordination Thresholds

Held, with a gap. More than thirty systems in one state within forty-eight hours clears the first threshold, a confirmed coordinated attack on two or more facilities within a single service area or regional cluster, on the first day by a defined criterion. Assessed: the threshold presumes geographic contiguity, and this event ran across communities sharing a device population and connectivity pattern. R-6.

6.7 §9.4 Recovery Sequencing, Phase 2

Held, with a gap. No automated control is restored until integrity is confirmed. Where E-1 has been executed, restoring automation from the resident configuration reinstalls the adversary's logic; the gate prevents a successful manual response from being undone. Assessed as the most operationally consequential provision for this event class; integrity confirmation should mean comparison against an offline gold copy at module level. R-3.

6.8 §9.1 Threat Detection and Indicator Framework

Held, with a gap. Several imminent-threat indicators map to the event, but the set is oriented to access and network state, with no configuration-integrity indicator, and against A-3 it is

structurally weak. The one organization that found modified project files did so by comparing its own sites, which a single-plant system cannot do; for the small system the offline copy of R-3 is the only configuration-integrity detection available. F-3, and F-6 sharpened.

6.9 Consequence Thresholds and the Pressure-Loss Pathway

Missed. The framework names distribution pressure as a critical function (§4.3), and Clayton County's boil water advisory is the established protective action. It does not trace pressure loss forward as a contamination initiator under adversarial conditions, where the loss of control may persist, recur at the adversary's election or be timed. No contamination has been reported; the pathway is stated by federal authority. The framework has the function and lacks the chain. R-8.

6.10 Correspondence With Federal Protective Guidance

Held, with a gap. The federal measures correspond closely and independently to §5.1, §5.2 and §9.4, the key switch naming program modification as the framework does not. End-of-life planning and cellular field connectivity as a distinct architecture have no counterpart; both lie in the installed base, a blind spot of a framework written from the process outward. R-9.

6.11 Summary of Design Basis Performance

Of the ten readings, three Held (Sections 6.1, 6.2 and 6.4), six Held, with a gap, and one, the pressure-loss chain, was Missed. Prevention and response are strong; detection is weak, in a way the military action extension does not remedy, since a compliant utility would probably have detected the access and not reliably the effect; consequence is incompletely traced. The asymmetry is structural: the cyber requirements were drawn from Oldsmar, where indication was never in question, and the lesson reached detection independence, not indication integrity.

7. Findings

Section references are to the DBA-W Sector Framework v1.5. The framework took in R-1 to R-10 at its v1.6; F-5, disposed to OI-2, it has not taken up.

7.1 F-1 — Indication Integrity Is Absent From the Design Basis

The Design Basis requires that operator visibility be preserved, not that what the operator sees be true or verifiable independently of the control system. §4.2.2 and §4.3 treat visibility as present or absent; E-2 is corrupted visibility, and the manual backup of §5.2, credited as an independent layer, consumes the same falsified indication. A layer that shares a compromised input with the layer it backs up is not a second layer.

Affected: §4.2.2, §4.3, §5.2, §9.4. Disposition: R-1.

7.2 F-2 — The Defense-in-Depth Principle Lacks a Common-Substrate Test

§4.1 counts measures without testing whether they share a substrate whose defeat removes them together. E-1 and E-2 defeated automatic protection and operator detection through one artifact: a facility satisfying §4.1 had a minimal defeat cutset of one. The deficiency is general to any functions in a common artifact, on common hardware or from a common configuration source. DBA-MA-DiD (*Defense-in-Depth Under Military Action*) supplies the barrier-independence ledger, the minimal substrate-defeat cutset and the independent-protection adequacy criterion, to be inherited, not originated.

Affected: §4.1. Disposition: R-2.

7.3 F-3 — The Engineering Workstation, Vendor Engineering Software and Controller Project File Are Not Design Basis Objects

The Design Basis protects parameters, functions and boundaries, not the artifact from which control behavior derives, and the campaign runs through it: A-3 uses the software, the exfiltration takes the files, E-1 modifies them. A compliant utility therefore lacks an offline authoritative copy of controller logic, authorization control over engineering sessions, and module-level change detection; without the first, integrity confirmation under §9.4 is inspection, not verification.

Affected: §4.2.2, §5.2, §9.1, §9.4. Disposition: R-3.

7.4 F-4 — The Threat Actor Taxonomy Does Not Accommodate Capability Proliferation

§7.5.1 puts capability and direction on one scale and assigns Tier 3 to §5, so it cannot represent state capability without state direction. The techniques are public, the condition unpatchable and the tooling commercial. Without state discipline, state capability is more likely to produce unintended consequence: descending tier raises consequence variance at constant capability. And a facility deferring under §7.1 relies on a §5 scoped to a Tier 3 that no longer describes Tier 3, connecting F-4 to F-5.

Affected: §7.5.1, §5 by reference. Disposition: R-4.

7.5 F-5 — The Applicability Determination Trigger Now Fires Without Discrimination

§7.1.3 requires re-determination of a deferral on a confirmed attack on a same-class facility in a comparable jurisdiction. Braham, a stable interior jurisdiction of approximately 1,700 with no evident strategic value under §7.1.2, is the paradigm of a geolocation-based deferral under §7.1.1, and it was attacked; the trigger now fires for effectively every community water system in the continental United States. Geolocation discriminates for the kinetic and denial modalities,

not for the cyber modality, whose reach is substantially geography-independent. Assessed remedy, not adopted: a modality-aware test, geolocation primary for kinetic and denial, geolocation and strategic value co-equal for cyber.

Affected: §7.1.1, §7.1.2, §7.1.3. Disposition: OI-2, not R.

7.6 F-6 — The Scope Threshold Excludes a Population That Is Materially Exposed

§1 scopes the framework to systems serving 3,300 or more. Braham serves approximately 1,700, the federal warning drew no line by size, and the structural exposure lies in the small-system population. The assertion of v0.2 that the adversary selects for the sub-threshold segment is withdrawn (Section 9.8): the campaign reached every size in the same period. What stands is that the framework is scoped away from a population materially exposed, without the resources for the full Design Basis or the peer comparison by which the decisive effect was found elsewhere. Full scope is not the remedy, since a standard that cannot be met is not adopted. Assessed remedy: a bounded minimum set derived from the Design Basis — no controller reachable from the public internet; the mode switch set against remote program modification; an offline authoritative copy of controller logic; an exercised manual procedure; and independent local indication with a periodic cross-check — five provisions needing no specialist staff or recurring license cost.

Affected: §1, and a new instrument. Disposition: R-5 and OI-3.

7.7 F-7 — The Framework Depends on a Single Cyber Reference Event

Oldsmar carries the whole cyber reference load, at §4.2.2, §5.3 and §7.4: single-facility, interactive, visible, detected by an operator who happened to see it, indication truthful throughout, each property inverted here. The lesson stopped at detection independence, and F-1 follows. Much of §7.4's constructed analog is now documented; marking which elements have moved from postulate to observation costs only the annotation.

Affected: §4.2.2, §5.3, §7.4. Disposition: R-7.

7.8 F-8 — Common Third-Party Configuration Is an Unaddressed Sector-Level Common Substrate

The Design Basis addresses the facility, not the shared integrator, service provider or reference architecture whose replication makes one weakness reachable at many utilities at once, as the announcement records may happen. This is F-2 at fleet scale: an architecture replicated across n utilities is one configuration whose defeat reaches all n, with a substrate outside the facility boundary. A utility's compliance is not established by its own architecture where that architecture was supplied to many; the §9.3 threshold does not reach utilities related only by

installer; and a small system cannot know its configuration is shared. Assessed: thirty systems in a weekend may be one architecture understood once and applied thirty times.

Affected: §4.1, §5.1, §9.3, and a scope element absent from §1. Disposition: R-6 as extended, and R-10.

8. Recommendation Register

Priority is on the series' one-to-three scale, 1 the most consequential, and gives the order of working. All ten dispose to the DBA-W Sector Framework and were taken in at its v1.6; the second part of R-7, the annotation of §7.4, is not in the framework at v1.15 and is reported to Project 9 in the batch 24 note.

ID	Recommendation	Target	Nature of change	Pri
R-1	Establish indication integrity as a required function	§4.2.2, §4.3, §5.2	Indication verifiable independently of the control path, by local instrumentation with a periodic operator cross-check; manual backup credited only where indication integrity is established.	1
R-2	Inherit the barrier-independence construct	§4.1	Defense in depth as an independence test, incorporating the ledger, cutset and adequacy criterion of DBA-MA-DiD.	1
R-3	Make the configuration artifact and its tooling Design Basis objects	§4.2.2, §5.2, §9.1, §9.4	Offline authoritative copy of logic; authorization of engineering sessions; module-level change detection; §9.4 integrity confirmed against the offline copy; program modification named in interlock independence.	1
R-4	Separate capability from direction	§7.5.1	Assess each independently; proliferated state capability is not bounded by tier.	2
R-5	Establish a sub-threshold minimum requirement set	§1, new instrument	Recognize systems below the threshold; a bounded minimum set as a short instrument derived from the Design Basis.	2
R-6	Extend the coordinated-attack threshold beyond geographic contiguity	§9.3	Trigger on a common device population, connectivity pattern, vendor platform, or integrator or service provider, independent of service area.	2
R-7	Introduce the second cyber reference event and annotate the first	§4.2.2, §5.3, §7.4	Add this event, retaining Oldsmar; annotate in §7.4 which elements of the analog have since been observed.	3
R-8	Trace the pressure-loss contamination chain under adversarial duration	§4.2, §4.3	Pressure loss as a contamination initiator, evaluated where loss of control persists, recurs or is timed.	1
R-9	Address the installed base and field connectivity	§4.2.2, §5.1	End-of-life forecast, tracking, replacement or isolation with decommission dates; carrier-network field connectivity as a distinct architecture.	2
R-	Establish shared third-party	§4.1, §5.1, §1	Assess architecture for elements supplied as a	2

10	configuration as a Design Basis consideration		common pattern to many utilities; state the obligation on integrators and service providers.	

9. Open Items

9.1 OI-1 — Re-Examination Conditions for the Tier Determination

Re-examined on evidence of an access-denial component — coordinated physical action, interdiction of repair or mutual aid, or timing to a condition that independently prevented access — or on recurrence where a denial mechanism is present. Official attribution to a state actor is not by itself a condition: it establishes direction, and the determination turns on modality. Re-examined at v0.3 against the expansion and the realized consequence: scale supplies no denial, and flooding at a wastewater facility is a process consequence, not an adversary preventing access. The determination stands.

9.2 OI-2 — Re-Scoping of the Applicability Determination

The modality-aware test of F-5 is recorded and not adopted; it changes §7.1 across the sector, bears on small-system adoption cost, interacts with R-4 and R-5, and needs its own review. The framework at v1.15 has not taken it up. It is the author's.

9.3 OI-3 — Instrument Form of the Sub-Threshold Requirement Set

Annex, standalone DBA-W instrument, or outreach product: a series architecture decision. The framework at v1.15 and the five facility classes name it a forthcoming DBA-W instrument.

9.4 OI-4 — Selection of the Initiating Event for DBA-W-UC-1

F-1 offers a companion initiating event to the first water use case's area denial: falsified indication with protection disabled. Open, and Project 9's.

9.5 OI-5 — Recording of the Reference Event Instrument Type

Closed at v0.4: the type is recorded in the Project 6 document architecture (Project6_Instructions v1.4, 2026-08-13).

9.6 OI-6 — Designator Reconciliation With the Oldsmar Reference Event

Closed at v0.5, and closed by DBA-W-RE-2 (*Oldsmar Water Treatment*) §9.1: designators number instruments by order of issuance, this instrument keeps DBA-W-RE-1, and Oldsmar is DBA-W-RE-2.

9.7 OI-7 — Version Reconciliation Affecting This Document's References

Two cache errors were found at v0.1. DBA Series Gaps and Recommendations cited the framework at a v1.6 that did not then exist, since overtaken by the issue of that version. TAF-W (*AI Quality Assurance Water Sector Application Guide*) cites the framework at v1.1, of consequence because §5.5 has since been revised; on the master read at migration it still does. This instrument's section references resolve against v1.5.

9.8 OI-8 — Corrections Applied to the Prior Version of This Document

Two statements of v0.2 were corrected at v0.3 and are recorded here rather than silently amended, because an instrument unwilling to apply its standard to itself has no standing to apply it to anything else. The access condition was misattributed: v0.2 identified CVE-2021-22681 as the access condition for the events characterized, and the announcement of 30 July establishes that the wave beginning 27 July targeted a different device family by a method for which no vulnerability identifier is cited. Section 3.3 now separates the two. The conclusion that architecture is the only control survives, resting on end-of-life hardware rather than an unpatchable flaw. The correction was later corroborated independently (Section 2.4). And the selection claim in F-6 was withdrawn (Section 7.6). Both errors share a cause: an assessment anchored on the first-reported subset of an event inherits that subset's properties.

10. Standing of This Document

10.1 What This Instrument Establishes

A characterized event record for citation; no requirement, and no precedent binding another event's characterization.

10.2 Relationship to the Oldsmar Reference Event

Oldsmar (DBA-W-RE-2) and this event are complementary: Oldsmar establishes that protection must not depend on operator detection; this event, that it must not depend on the integrity of indication or the availability of remediation. DBA-ESS-RE-2 (*Multistate 911 Outage*) cites F-2, which is kept.

10.3 Maintenance

Revised on the federal investigation's completion or reporting; on official attribution, which would alter Sections 2.3 and 3.6 and not the tier determination; on material expansion of the confirmed scope; or on a condition of OI-1. Whether the activity reported in other states in late July 2026 belongs to this campaign is held in Project 9's register; an independent source establishing campaign membership would mean a scope revision or a new instrument, the author's choice.

References

The sources named in Section 2.1, the predecessor keeping the full list; CVE-2021-22681 and Rockwell Automation advisory SD1672; the infracritical playbook cited at Section 2.4 only; the DBA-W Sector Framework v1.5; DBA-W-RE-2; DBA-MA-DiD; DBA Series Gaps and Recommendations; TAF-W; CB-Framework_Cross_Sector (*Command Broker Framework, Cross-Sector*), cited by designation against CB-REG; AWWA J100-21; America's Water Infrastructure Act of 2018; the Foundational Definitions named in Section 1.2, with FD-REG; and the DBA-MA Parent Framework.

Revision History

Version 0.1, July 2026: initial draft; the instrument type and its distinction from the use case; three access and two effect elements; military action tier classification found against on the absence of denial; seven findings, recommendations and open items, against the framework at v1.5. Version 0.2, July 2026: the element register added to Section 4.

Version 0.3, August 2026: on the expansion to at least twelve states and the announcement of 30 July 2026, Section 3.1a new; Section 3.3 separates two access conditions, end-of-life hardware governing; A-4, C-1, Sections 6.9 and 6.10, and F-8 new; F-6's selection assertion withdrawn; R-8 to R-10 added, R-6 extended; OI-1 re-examined, the determination standing; OI-8 records the corrections.

Version 0.4, August 2026: OI-5 closed. Version 0.5, August 2026: OI-6 closed, Oldsmar instrumented as DBA-W-RE-2. Version 0.6, August 2026: Section 2.3 records the CyberAv3ngers claim as reported, not an official attribution; Section 3.1a takes the scope of the analysis of 18 August 2026; Section 5.1 notes the warning-intent claim; tier determination and findings unchanged. Version 0.7, September 2026: Foundational Definitions cited by designation under the citation-form determination, two tokens having gone stale on 3 September; no finding changed.

Version 0.8, 13 October 2026: migrated to Project 11 under the determination of 9 October 2026 (Reference Events Move to Their Own Project), compressed under the Reference Event Compression Standard v0.3 and brought to house style. On the triage, a hard cut: the geopolitical frame, advisory context and federal response each to a paragraph, Section 6 to the four verdicts, its ten readings keeping their numbers. Section 2.4 records the external corroboration of the v0.3 correction, as Project 9 determined on 8 September 2026. The §9.3 thresholds are called coordination thresholds and R-6 retitled accordingly. Open items and history are put in order, the history in prose. TAF-W and CB-Framework_Cross_Sector are cited by designation. Unchanged: the designator, title, tier determination, attribution statements, Section 3.1a's account of scope, and all eight findings and ten recommendations in number and substance. Verified: every finding and recommendation number present at v0.7 is present.